

MUHAMMAD ARSLAN AKHTAR

Elite Offensive Security Researcher | Identity Access Management SSO Specialist (Zendesk)

Bridging Offensive Intelligence, AI Architecture, and Enterprise Risk Governance

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PROFESSIONAL SUMMARY

I have built one of the largest known empirical datasets of Zendesk IAM & SSO misconfiguration patterns in the information security community over the past 6 years, working as an Elite Offensive Security Researcher focused on Identity Access Management Vulnerabilities. I have engaged 500+ live production deployments, assessed their business logic vulnerabilities and documented 40+ distinct Broken Access Control & SSO Authentication Bypass patterns that commercially available scanning tools cannot detect. This work earned me a Top 1% spot globally with a 99th percentile Impact score on HackerOne.

My strategic approach for advisory-first methodology frames technical findings within the compliance language of ISO 27001, GDPR, NIST, and OWASP for C-suite audiences has commanded advisory fees of up to \$25,000 per engagement and forged verified relationships with executive leadership (CEOs, CTOs, CISOs, VPs, and Directors) at over 500 global SaaS organisations, including Adobe, Instacart, CompTIA, and, Wrike Inc. Notably, after two engagements with Hootsuite led me to be recruited directly by Dr Cuneyt Karul (VP IT & Security), who described my advisory report as one of the most exemplary reports he had ever encountered later used as a learning guidance for students at the University of Waterloo.

I am currently working on ZenGuard, a proprietary AI-powered deployable adversarial reasoning engine trained on my 6 years collection '40+ detection rules' dataset. This engine will be serving as stepping stone and its success would trigger a technical foundation for PakCyberShield, I envision as South Asia's first indigenous managed BugBounty-as-a-Service (BBaaS) platform.

EMPLOYMENT HISTORY

Ethical Hacker & Elite Security Researcher **Global Platforms (Self-Employed)** 2020 – Present | Remote

- Validated Professional Record:** Delivered security assessments across private enterprise contracts and public platforms; successfully navigated 200+ NDAs for direct-to-client audits. Formal letters of appreciation from C-suite, Director, and VP-level stakeholders at named organisations globally.
- Zendesk Single Sign-On (SSO) Architecture Specialization:** Conducted systematic security assessments of Zendesk Suite deployments including Zendesk Support, Guide and Admin Centre across 500+ global enterprise organisations. Documented 40+ distinct SSO misconfiguration patterns spanning authentication bypasses, domain-trust gaps, email-domain identity federation failures, and Help Center access control misconfigurations. No commercial scanner or SSPM tool currently detects this vulnerability class assessment requires adversarial reasoning trained on proprietary empirical data accumulated over six years of active research.
- HackerOne & BugCrowd — Top 1% Impact (99th Percentile):** Ranked 99th percentile for Impact and 78th percentile for Signal (5.00). Specialist in SSO Authentication Bypass and Broken Access Control across enterprise-grade applications; responsibly disclosed 40+ high-severity vulnerabilities to Fortune 500 organisations across Cloud Imperium Games, Age of Learning, Cambly, and others. At BugCrowd, I have 91.3% submission accuracy with 21 valid vulnerabilities. Hall of Fame across 18 programmes, including Checkr, Fiverr, Zillow, Under Armour, and Edge Delta.

Information Security Consultant **Hootsuite Inc.** May 2024 – present | Remote, Canada

Reference: Dr Cuneyt Karul, PhD, VP IT & Security Operations

- Academic Recognition of Advisory Quality:** The vulnerability advisory submitted to Hootsuite was described by Dr Cuneyt Karul PhD, VP of IT & Security Operations, as one of the most exemplary security reports he had encountered — subsequently used as a learning document for his engineering team and as a teaching paper in his visiting lectures at the University of Waterloo. This engagement led directly to my formal appointment as Information Security Consultant.
- Offensive Security:** Executed full-scope penetration testing and vulnerability assessments on 300+ portfolio enterprise-grade customer-facing systems identified and neutralised 40+ high-impact vulnerabilities.
- Operational Improvement:** Partnered with DevSecOps teams to implement hardening strategies that **improved MTTR by 40%** without disrupting business availability.
- Cloud Compliance & Security Culture Leadership:** Secured CI/CD pipelines via automated container image scanning and secret detection, **reducing deployment-phase misconfigurations by 60% and ensuring 100% GDPR compliance**. Aligned controls with ISO 27001, NIST CSF, and OWASP Top 10. Scaled the Security Champions Programme across multiple engineering squads, fostering a Security-by-Design culture that reduced total vulnerability mitigation time by 60%.

Lead AI & Data Solution Architect **Premium Houseware UK Ltd.** March 2020 – Present | Remote UK

- Predictive Modelling & Inventory Optimization:** Designed and deployed time-series analytics, achieving a 65% improvement in forecasting accuracy and a 97% reduction in inventory stockouts.
- Revenue Growth & Agentic Workflows:** Engineered ML models to optimize price and product-bundling. Output: 100% Sales jump. Currently developing agentic AI workflows to automate complex cross-departmental operations.
- Operational Automation & CRM Integration:** Developed and integrated AI agents to automate CRM workflows, resulting in a 45% improvement in operational efficiency and a 50% reduction in recurring overhead costs.

PROJECTS & TECHNICAL INITIATIVES

- **ZenGuard — AI-Powered Zendesk SSO Misconfiguration Detection Engine:** Developing a proprietary AI SaaS product trained on 500+ real-world Zendesk vulnerability records and 40+ detection rule patterns. Adversarial reasoning engine simulates attacker cognition to identify SSO trust chain exploitability in real-time, generating CVSS-scored executive reports mapped to GDPR, OWASP Top 10, and ISO 27001. Designing to expand across Freshdesk, Okta, Salesforce Service Cloud, and Jira Service Management. It is the technical engine driving the success to trigger a bigger mission The PakCyberShield: South Asia’s first indigenously developed BugBounty-as-a-Service platform.
- **Cloud DevSecOps Pipeline Security:** Designed security controls within CI/CD pipelines (GitLab CI, Jenkins); integrated SAST, DAST, container image scanning, and secret detection — reducing misconfigurations by 60%, aligned with ISO 27001 and NIST CSF.
- **Azure Cloud Migration Security Consultancy:** Conducted risk assessments, identity access modelling, and secure architecture baselines for on-prem to Azure migration; implemented conditional access, Key Vault, NSGs, and Azure Monitor/Sentinel.
- **Threat Modelling & Secure SDLC (OWASP + STRIDE):** Led threat modelling workshops for Agile delivery teams using STRIDE and OWASP Threat Dragon; integrated countermeasures into product backlogs and CI/CD pipelines.
- **Docker Container Vulnerability Assessment:** Led evaluation of runtime threats in containerised microservices; authored formal whitepaper delivering mitigation roadmaps to senior architects.

EDUCATION

MBA — Management	IBMS, University of Agriculture Faisalabad (UAF)	2017 – 2020
2.5 years Master of Business Administration with distinction CGPA: 3.84 / 4.00, focused on Strategic Leadership, Technology Management, Data-Driven & Intelligent Decision Making, Management Information System. Capstone: Optimising business processes with data-driven insights.		
BS Honours — Applied Mathematics & Statistics	GC University Faisalabad (GCUF)	2013 – 2017
Mathematical Modelling, Statistical Analysis, Cryptography, Algorithms.		

CERTIFICATIONS & PROFESSIONAL TRAINING

Advanced Cybersecurity & Cloud Security

- **AI Cloud Security Residency (12-Month Intensive):** *Boston Institute of Analytics (BIA) & Newton AI Technologies.*
 - Completed **200+ hands-on labs** in offensive/defensive operations and AI-integrated threat defence.
 - **High-Volume Project Delivery:** Completed and delivered **12+ end-to-end projects** utilising advanced Linux environments and AI security frameworks, managing the full lifecycle from threat modelling to executive remediation.
 - **Cloud Governance:** Worked closely with the DevOps team to Automate Kubernetes security using GitLab CI and Policy-as-Code, reducing manual compliance overhead by **40%**.
 - **Specialized Research:** Authored the '*Docker Container Vulnerability Assessment*' whitepaper, delivering runtime threat mitigation roadmaps to senior architects.
- **AI in IT Operations - AIOPs (RQF Level 6):** *EduQual Level 6 AI-Nafi Programme (500+ Labs).*
 - **Core Technical Skills:** Ethical Hacking, **Vulnerability Assessment**, Network Penetration Testing, **Web Application Penetration Testing (WAPT)**, and **Cyber Threat Intelligence (CTI)**.
 - **Security Tooling & Ops:** Expert use of **Metasploit**, **Nessus**, and SIEM tools. Specialised in **Cloud Security**, Cloud Computing, and **DevOps/SysOps Strategies** for high-availability environments.
 - **AI & Management:** AI Project Management, **Machine Learning**, AI Ethics, and **Artificial Intelligence for Operations (AIOPs)**.
 - **Development & Automation:** **Python Programming**, **Bash Scripting**, Automation & Workflow Management, and **Secure Coding Practices**.

Artificial Intelligence & Data Science

- **Advanced Generative AI & Agentic Systems:** Specialised in **LLMs**, **Agentic AI**, and **Computer Vision**, engineered autonomous security response workflows.
- **Data Science & AI Engineering:** Intensive 4-month specialisation in **Python**, **ML**, **MLOps**, **Deep Learning**, and **NLP**.
- **Professional Data Analytics:** Advanced certification in **SQL**, **Power BI**, **GIS**, and **Excel**; focused on deep-dive data forensic analysis and visualisation.